

Petrel Geology Isochore and Isopach processing

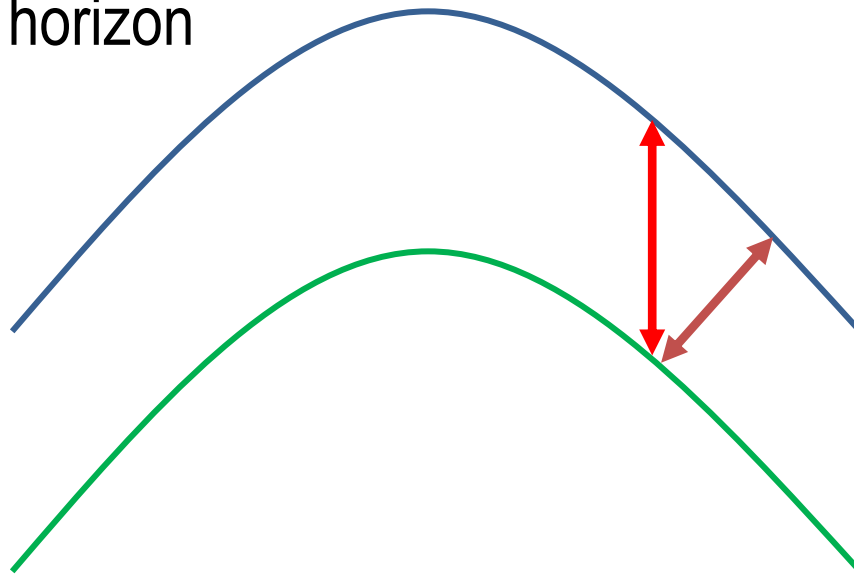
Learning objectives

After you complete this module, you will know how to:

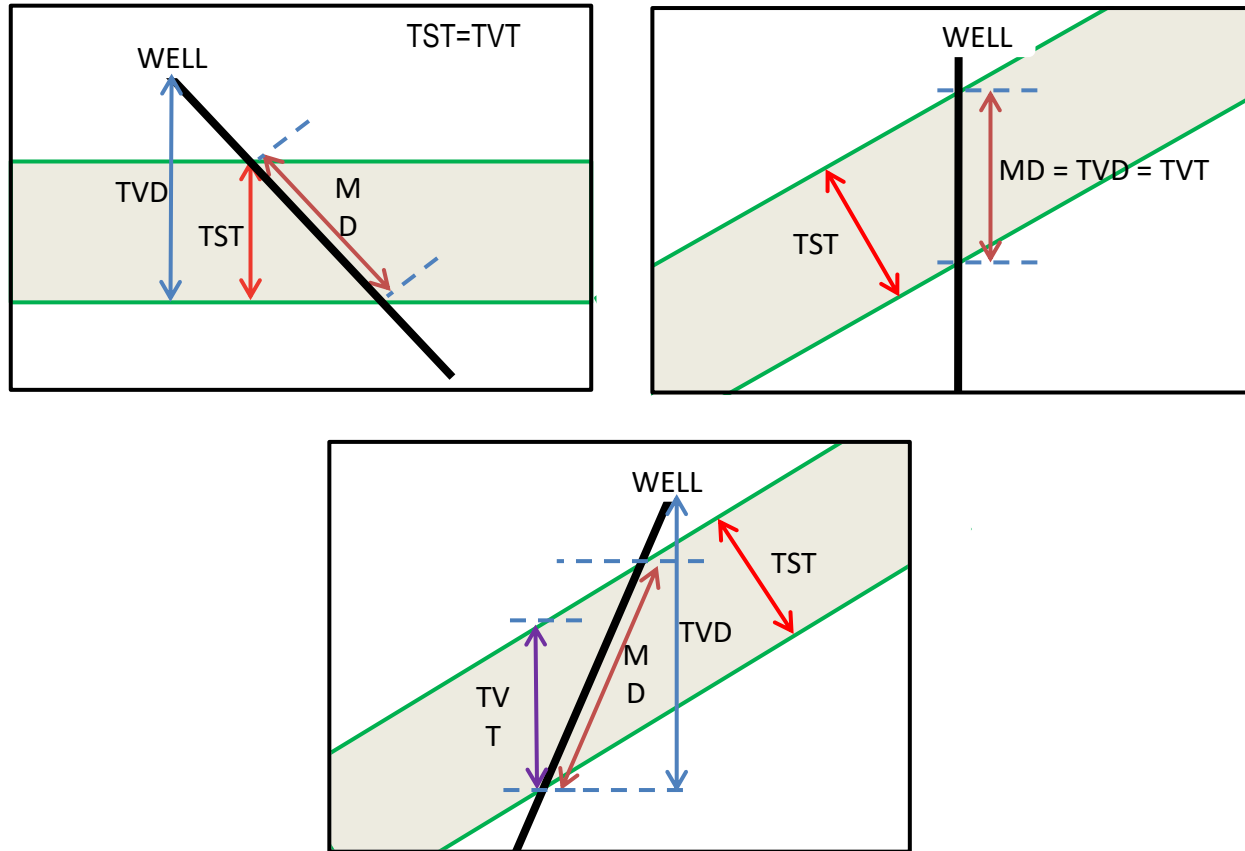
- define and differentiate between an isochore and isopach
- describe and define TVT, TST, and MD
- create isochore points from deviated and vertical wells
- create zone logs
- create well tops attributes

Isochore processing: Definitions

- **Isochore**: Thickness between two horizons, measured vertically
- **Isopach**: Thickness between two horizons, measured orthogonally to the horizon



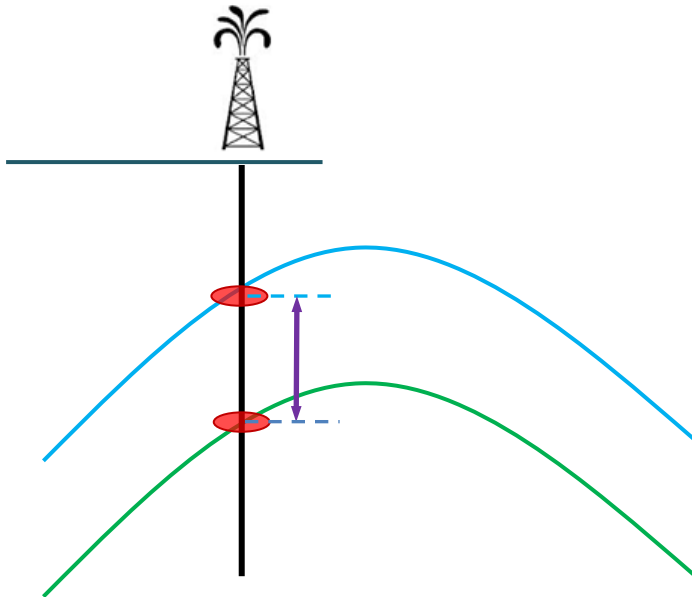
Definitions: TST, TVT, and MD



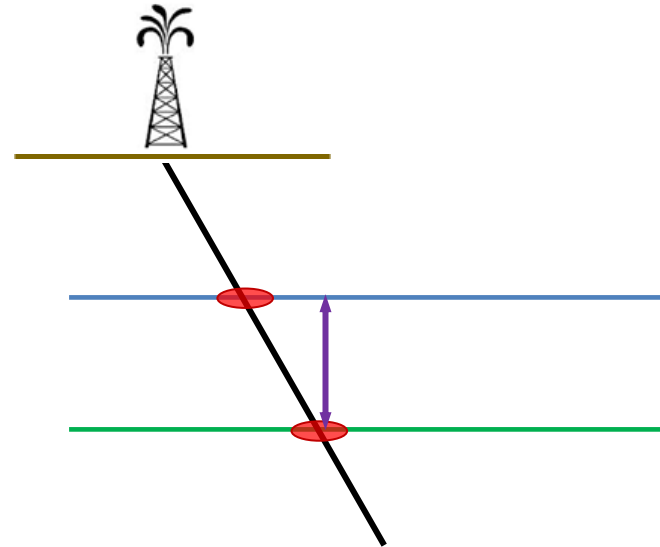
Calculate isochore values

Simple case: Thickness = TVT

Vertical well $Z_{\text{top}} - Z_{\text{base}}$

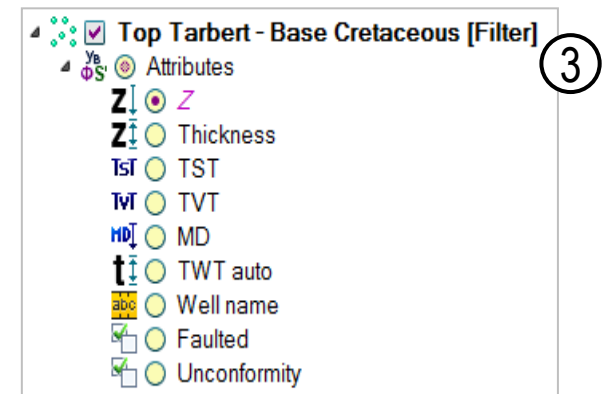
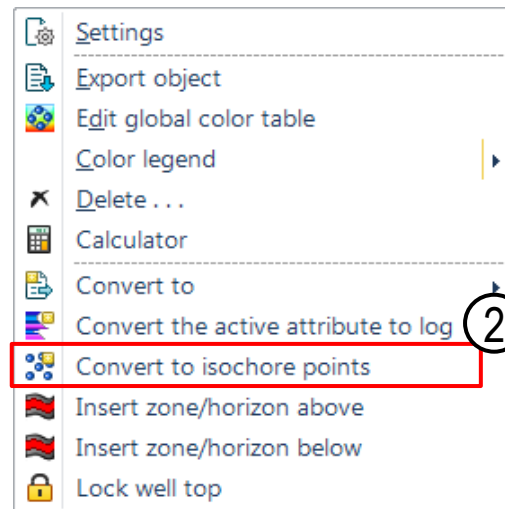
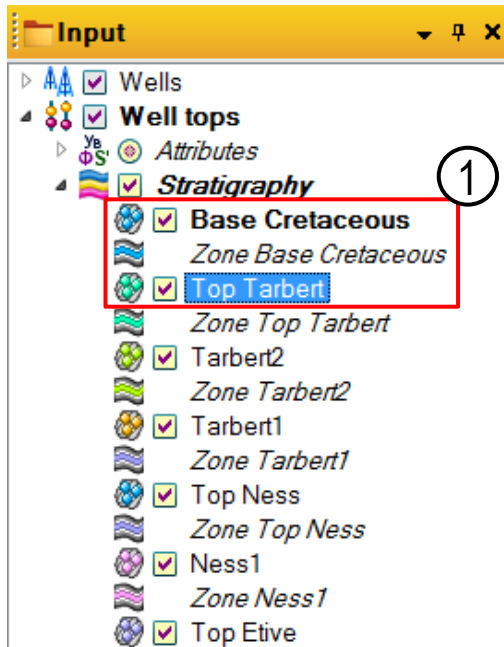


Deviated well and horizontal surface $Z_{\text{interpolated}} - Z_{\text{base}}$



Create isochore points (vertical wells)

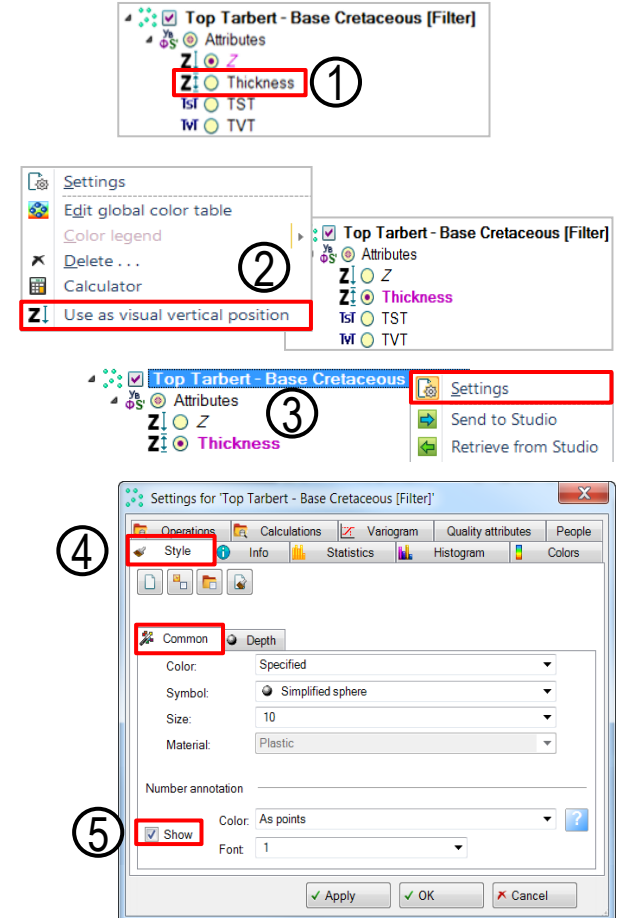
1. Highlight a well top and right-click another that will define the isochore interval.
2. Select *Convert to isochore points*.
New point set data is created in the **Input** pane.
3. Visualize the dataset in a **3D window**.



Visualize isochore points

1. Right-click the Thickness attribute.
2. Select *Use as visual vertical position*. The points now are shown as thickness points in a **3D window**, and they are marked in pink in the *Attributes* folder.
3. Right-click the isochore points to open the settings.
4. On the **Style** tab, click the **Common** subtab.
5. Select *Show* and click *OK*.

Display them in a **3D window**. Note that now they are above the wells, according to their visual vertical position (thickness depth).

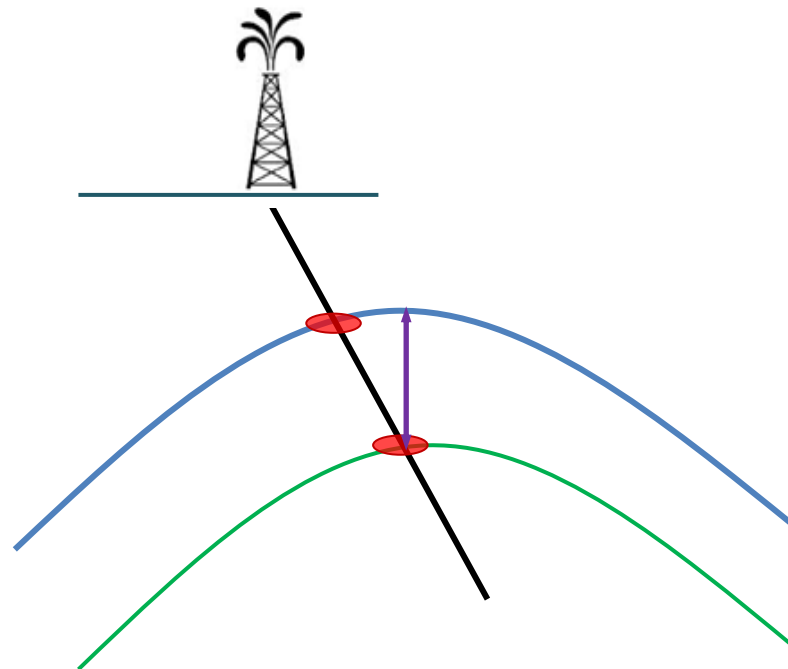


Calculate Isochore values

Complex case: Thickness $\neq$ TVT

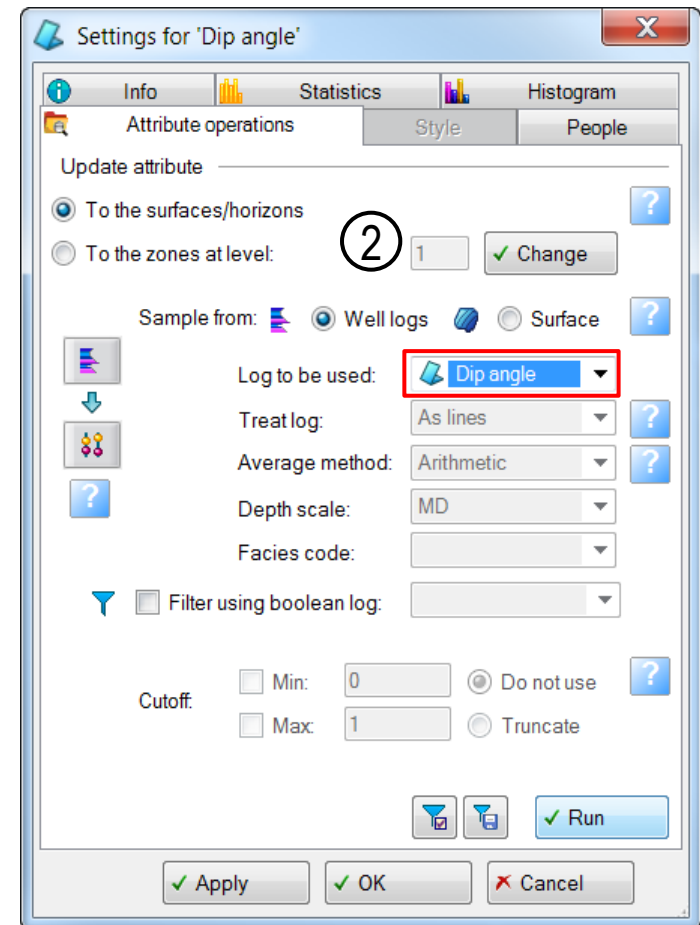
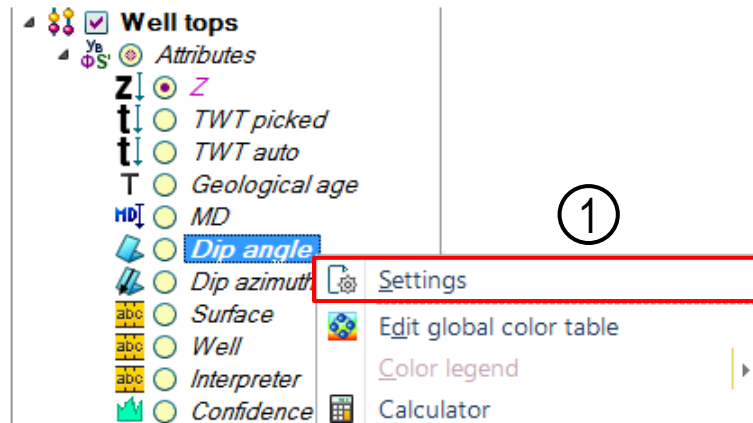
Deviated well and
deviated surface

$$Z_{\text{interpolated}} - Z_{\text{base}}$$



Extract Dip or Azimuth well tops attributes from well logs

1. Open the settings for the Dip Angle attribute.
2. On the **Attribute operations** tab, select to sample from Well logs, and choose Dip Angle as the log to use.



Extract TVT and TST from well logs

1. Open **Settings** for the *Wells* folder and click the **Thickness** tab.
2. Select *From logs* and select Dip Angle and Dip Azimuth.
3. Set a sample rate of 0.5.
4. Click *Run*.
5. Right-click the *Well tops* folder and click *Zones spreadsheet*. The TVT/TST values are updated.

The image displays three screenshots illustrating the process of extracting TVT and TST from well logs.

Screenshot 1: Settings for 'Wells' - Thickness tab

The 'Settings for Wells' dialog box is shown with the 'Thickness' tab selected. The 'True vertical/stratigraphical thickness (TVT/TST)' section is configured as follows:

- Source: **From logs** (selected)
- Dip: **DipAngle** (selected)
- Azimuth: **DipAzimuth** (selected)
- Sample rate: **0.5** (MD)

The 'Run' button is highlighted with a red box and a circled '4'.

Screenshot 2: Well tops folder context menu

The 'Well tops' folder is right-clicked, and the 'Zones spreadsheet' option is highlighted with a red box and a circled '5'.

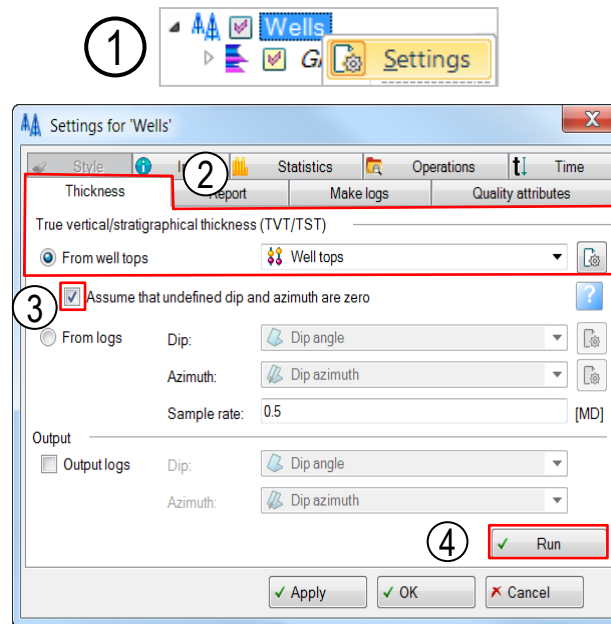
Screenshot 3: Well top zones spreadsheet for 'Well tops'

The 'Well top zones spreadsheet' window is shown, displaying a table with columns: Well, Start MD, Zone log, TVT zone, and TST zone. The TVT and TST columns are highlighted with a red box.

Well	Start MD	Zone log	TVT zone	TST zone
14 A10	1858.55	Zone Base Cretace	0.00	0.00
28 A10	1858.55	Zone Top Tarbert	0.00	0.00
29 A10	1858.57	Zone Top Ness	4.11	3.13
56 A10	1878.07	Zone Ness1	12.22	9.45
72 A10	1858.57	Zone Tarbert1	0.00	0.00
89 A10	1858.55	Zone Tarbert2	0.00	0.00
13 A15	1879.35	Zone Base Cretace	0.00	0.00
27 A15	1879.35	Zone Top Tarbert	1.37	0.90
30 A15	1922.59	Zone Top Ness	16.63	10.80
55 A15	1992.22	Zone Ness1	20.75	13.42
71 A15	1805.73	Zone Tarbert1	4.13	2.68
84 A15	1885.48	Zone Tarbert2	4.71	3.07
12 A16	1805.45	Zone Base Cretace	0.00	0.00
26 A16	1805.47	Zone Top Tarbert	1.40	0.95
31 A16	1875.86	Zone Top Ness	6.23	4.24
57 A16	1936.51	Zone Ness1	0.72	0.49
73 A16	1831.87	Zone Tarbert1	6.46	4.34
85 A16	1815.92	Zone Tarbert2	2.21	1.45
11 B2	2012.93	Zone Base Cretace	0.00	0.00
25 B2	2012.93	Zone Top Tarbert	1.53	0.99

Extract TVT and TST from well tops without dip and azimuth data (1)

1. Open the **Settings** for the *Wells* folder.
2. On **Thickness** tab, select to extract the TVT and TST from well tops.
3. Turn on *Assume that undefined dip and azimuth are zero*.
4. Click *Run*.



Extract TVT and TST from well tops without dip and azimuth data (2)

5. Check TVT and TST values in a Well top spreadsheet.
6. Observe that any edited well tops are updated automatically in the Well top spreadsheet, Zone spreadsheet, and also in Zone logs.

Well top spreadsheet for 'Well tops 1' ⑤

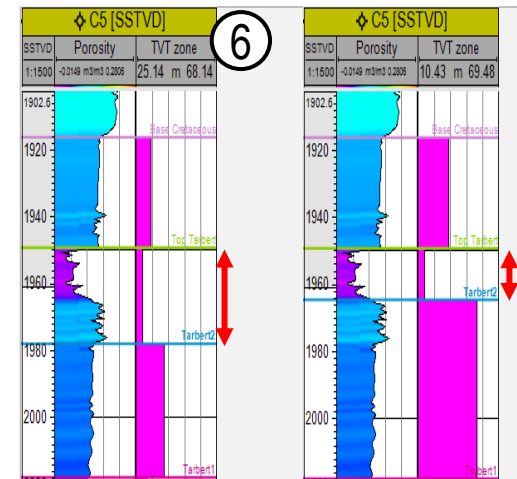
Well identifier	Surface	TVT	TST
8	C5	Base Cretaceous	1916.14
17	C5	Top Tarbert	1949.22
40	C5	Tarbert2	1979.11
51	C5	Tarbert1	2017.77
66	C5	Top Ness	2048.51
79	C5	Ness1	2102.05
96	C5	Top Etive	2166.61

Well top spreadsheet

Well top zones spreadsheet for 'Well tops 1'

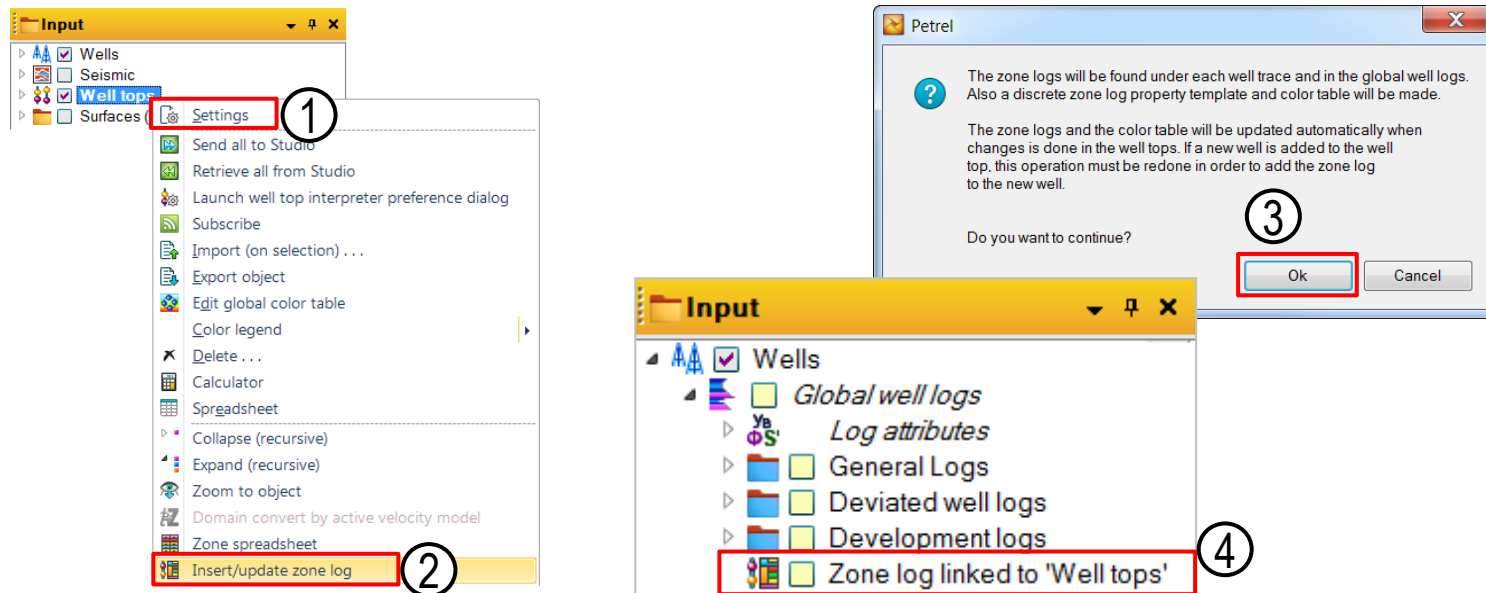
Well	Start MD	Zone log	TVT zone	TST zone
8	C5	1919.01 Zone Base Cretac	33.08	33.08
17	C5	1952.52 Zone Top Tarbert	29.90	29.90
40	C5	1990.36 Zone Tarbert2	38.66	38.66
51	C5	2041.03 Zone Tarbert1	30.74	30.74
66	C5	2074.78 Zone Top Ness	53.54	53.54
79	C5	2129.56 Zone Ness1	64.56	64.56

Well top zones spreadsheet



Create zone logs

1. Right-click Well tops to open its settings.
2. Select *Insert/update zone log*.
3. Click OK in the message window.
4. The new Zone log is stored under the **Wells** folder inside the **Global well logs** folder.



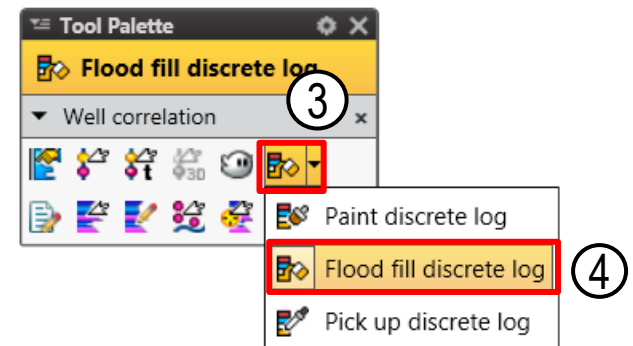
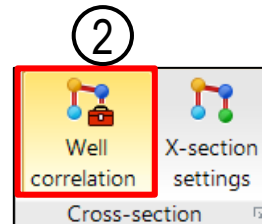
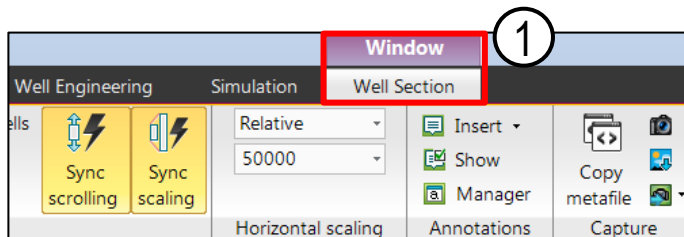
Edit zone logs (1)

There are two ways to edit the Zone logs:

- Automatic update from a right-click  Insert/update zone log
- Interactive update from a Flood fill discrete log

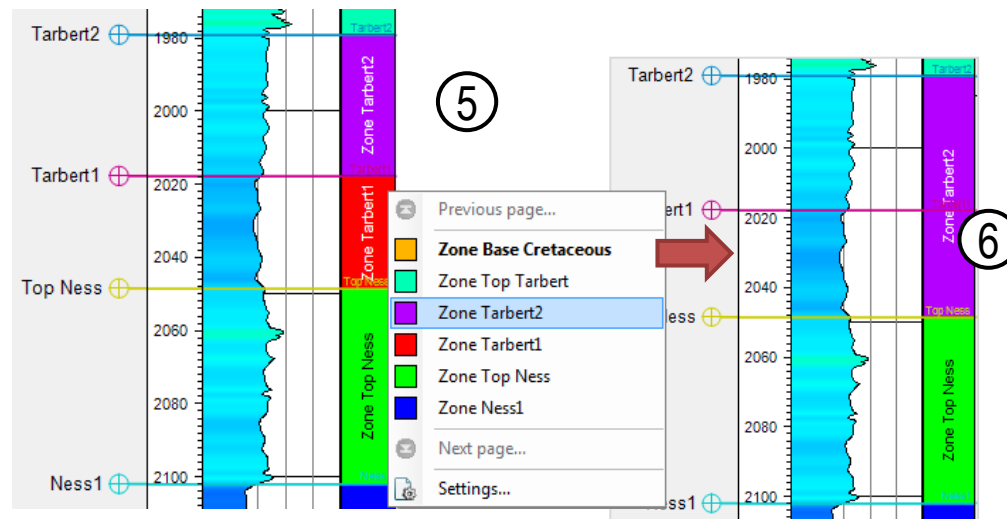
To edit the zone log using the second method:

1. Open the **Well section** tab.
2. In the **Cross-section** group, click *Well correlation*.
3. On the **Tool Palette**, click *Paint discrete log*.
4. Click *Flood fill discrete log*.



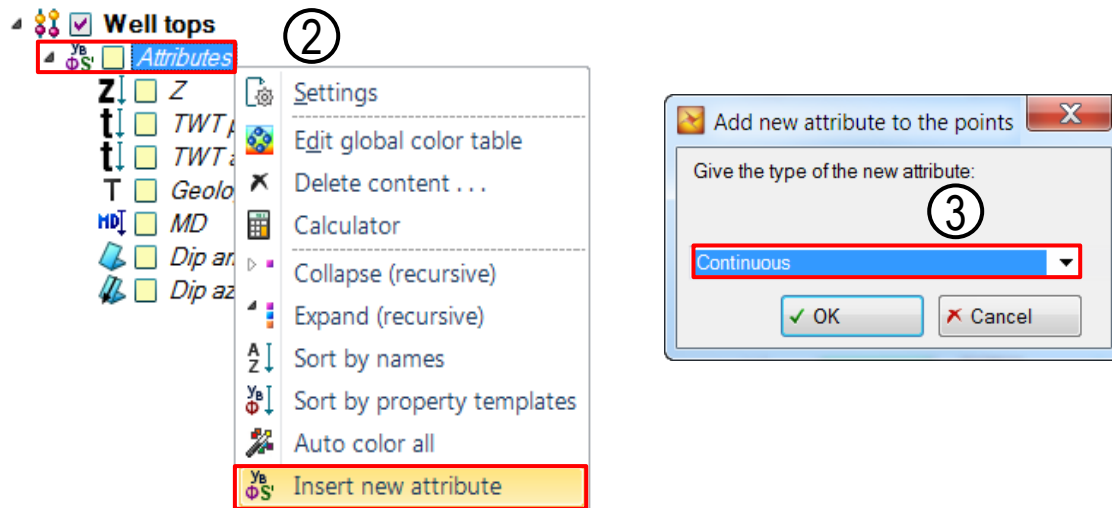
Edit zone logs (2)

5. Right-click the Zone log track and select the new color for the new zone code.
6. Click the interval you want to change in the Zone log track.



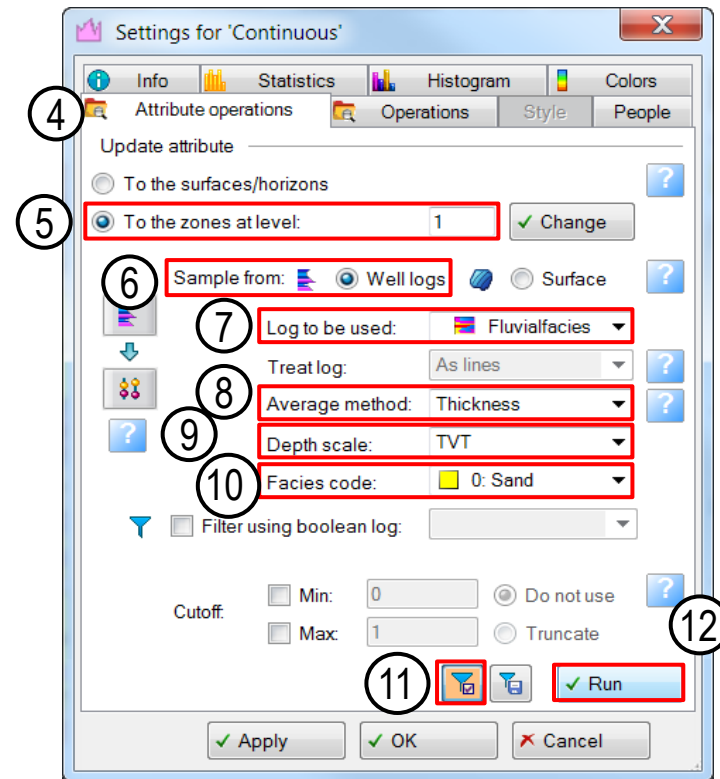
Create Well tops attributes (1)

1. Display the wells, the facies data, and the top markers for the intervals of interest in a **Well section** window.
2. In the **Well tops** folder, right-click the **Attributes** subfolder and click *Insert new attribute*.
3. In the dialog box, set the attribute type to Continuous and click *OK*.



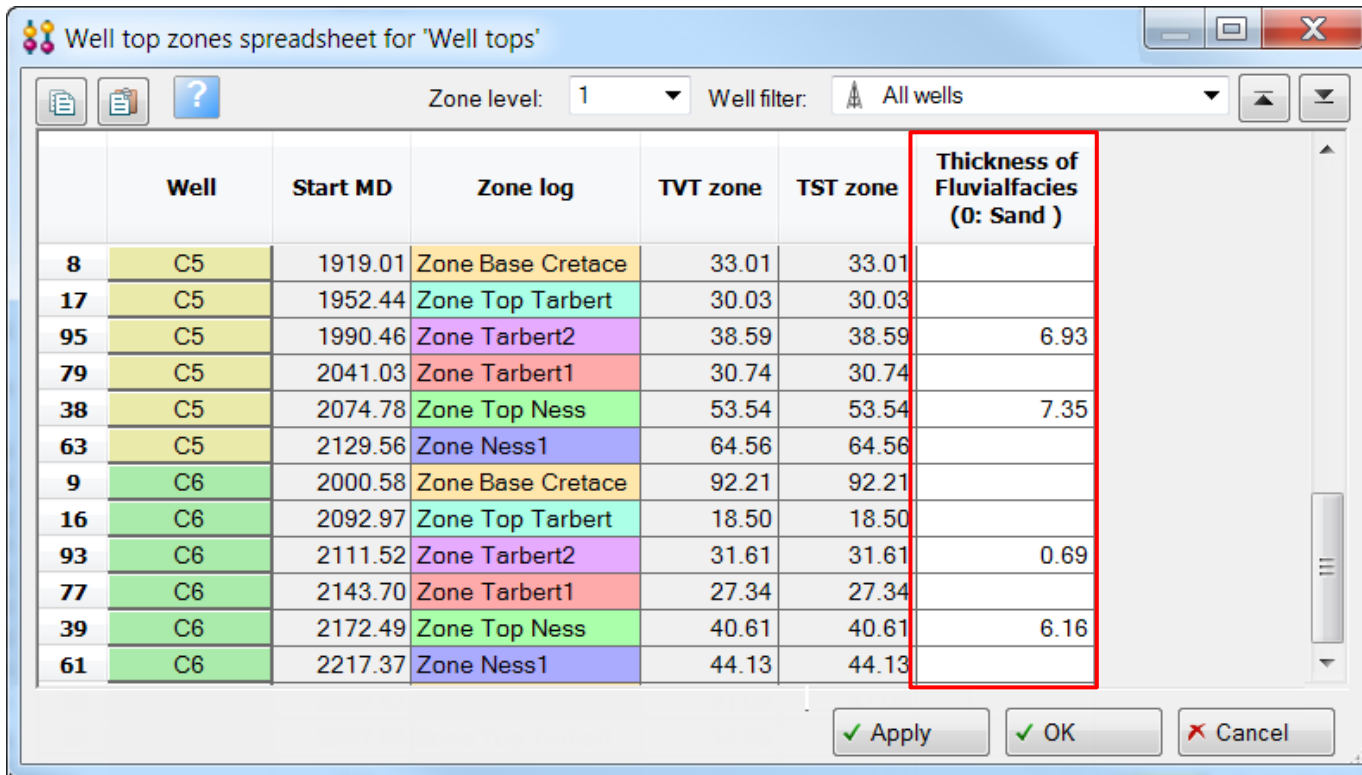
Create Well tops attributes (2)

4. In the **Settings** dialog box for the Continuous attribute, click the **Attribute operations** tab.
5. Turn on *To the zones at level* and set it to 1.
6. In the Sample from field, select *Well logs*.
7. In the Logs to be used field, select the facies log that you want to use.
8. In the Average method field, select Thickness.
9. In the Depth scale field, select TVT.
10. In the Facies code field, select the facies code that you want to extract.
11. Click *Apply the calculation using a filter based on well tops selected in the input tree (Filtered by yellow sub icons)*.
12. Click *Run* to perform the calculation or *OK* to run and close the dialog box.



Create Well tops attributes (3)

13. Check the values in the Well top zones spreadsheet.



Well top zones spreadsheet for 'Well tops'

Zone level: 1 Well filter: All wells

	Well	Start MD	Zone log	TVT zone	TST zone	Thickness of Fluvialfacies (0: Sand)
8	C5	1919.01	Zone Base Cretace	33.01	33.01	
17	C5	1952.44	Zone Top Tarbert	30.03	30.03	
95	C5	1990.46	Zone Tarbert2	38.59	38.59	6.93
79	C5	2041.03	Zone Tarbert1	30.74	30.74	
38	C5	2074.78	Zone Top Ness	53.54	53.54	7.35
63	C5	2129.56	Zone Ness1	64.56	64.56	
9	C6	2000.58	Zone Base Cretace	92.21	92.21	
16	C6	2092.97	Zone Top Tarbert	18.50	18.50	
93	C6	2111.52	Zone Tarbert2	31.61	31.61	0.69
77	C6	2143.70	Zone Tarbert1	27.34	27.34	
39	C6	2172.49	Zone Top Ness	40.61	40.61	6.16
61	C6	2217.37	Zone Ness1	44.13	44.13	

Apply OK Cancel

Summary

In this module, you learned about:

- creating isochore points
- handling vertical and deviated well situations to extract TVT and TST
- extracting TVT and TST from Well tops without Dip and Azimuth
- creating Zone logs and Attributes from Well Tops

Exercises

Create and process isochores.